

Low Power Design Techniques for Single-Bit Audio Continuous-Time Delta Sigma ADCs Using FIR Feedback

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Abstract—Single-bit continuous-time delta-sigma modulators (CTDSM) using FIR feedback DACs inherit the appealing aspects of both single-bit and multibit designs, without the disadvantage of either approaches. In this work, we give a method for stabilizing a CTDSM that uses an FIR feedback DAC. Further, we show that increasing the number of taps beyond a certain number (dependent on the architecture and oversampling ratio of the modulator) does not improve performance. The results of our analysis are incorporated in the design of a third-order audio CTDSM which achieves a peak A-weighted SNR of 102.3 dB (raw SNR of 98.9 dB) and a spurious-free dynamic range of 106 dB in a 24 kHz bandwidth, while consuming only 280 μ W from a 1.8 V supply.

Index Terms—Active-RC, analog-to-digital conversion, compensation, continuous-time, FIR DAC, integrator, oversampling, Sigma-Delta.

I. INTRODUCTION

HIGH resolution analog-to-digital converters for audio applications have traditionally been realized as multibit delta sigma modulators, employing switched capacitor circuitry. However, continuous-time $\Delta\Sigma$ modulators (CTDSM) are becoming increasingly popular even in these low bandwidth applications due to their low power consumption and the ease with which they can be driven. Most high performance audio CTDSMs [1]–[3] which have achieved linearity and noise comparable to a discrete-time design, however, have used multibit quantizers.

Using a multi-level quantizer has several advantages. First, the increased number of levels enables a lower over sampling ratio (OSR) to achieve a desired signal to quantization noise ratio (SQNR). The reduced step size in the feedback waveform reduces the sensitivity of the modulator to clock jitter, as well as the slew rates required of the opamps in the loop filter. However, the complexity of the ADC used in the quantizer increases exponentially with the number of bits. Even though the comparators in the quantizer lend themselves to low power operation,

clock generation and distribution can consume significant current. This, unfortunately, is difficult to estimate during the architectural design phase. Moreover, mismatch in the unit elements of the feedback DAC degrade the inband SNDR of the modulator, necessitating some form of mismatch correction, like calibration or dynamic element matching (DEM). This further increases the power dissipation and design time of the quantizer. In contrast, using a single-bit quantizer, where the feedback DAC is inherently linear, dramatically simplifies the quantizer design. However, the full-scale two-level feedback waveform places increased demands on the linearity of the loop filter, as well as increases the sensitivity of the modulator to clock jitter.

From the discussion above, it is seen that a multibit loop complicates the quantizer design at the expense of a simplified loop filter. The opposite is true in a single-bit modulator. Recognizing this, several recent works have attempted to alleviate the linearity and clock jitter problems associated with a single-bit design. [4] and [5] use integrators based on opamp assistance, which is a circuit technique that enhances the linearity of an opamp in a power efficient manner. [6] uses a feedback DAC which incorporates a finite impulse response filter (FIR-DAC). The FIR DAC principle is not new [7], [8] – however, relatively few designs seem to have exploited this technique. The basic idea behind the FIR DAC is to feedback a filtered version of the single-bit quantizer output, as shown in Fig. 1(a). Due to the high frequency attenuation of the FIR filter, the DAC output is a multi-level waveform, like in a multibit quantizer. Thanks to this, the FIR-DAC approach has low clock jitter sensitivity and relaxes the linearity requirements of the loop filter. In practice, the filter and DAC combination are implemented in a semi-digital fashion, as shown in Fig. 1(b), which makes the FIR DAC inherently linear in spite of mismatch. Due to the single-bit quantizer, the ADC design is simple and consumes very little power. A modulator employing a single-bit quantizer and an FIR DAC, therefore, combines the best features of single-bit and multibit operation. The aim of this work is to investigate the possibility of achieving a dynamic range in excess of 100 dB, and a correspondingly excellent linearity using a single-bit CTDSM with an FIR feedback DAC.

FIR feedback presents several design challenges. Due to the delayed nature of the feedback waveform, the modulator needs to be carefully stabilized to restore the noise transfer function (NTF). An issue particularly relevant to a continuous-time oversampled converter based on an FIR filter is the robustness of loop compensation in the face of time constant variations in the

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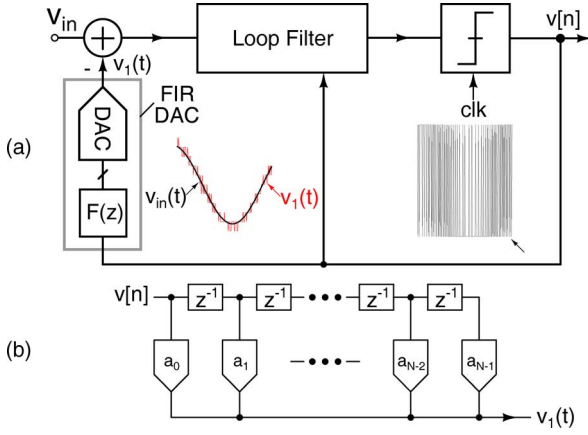


Fig. 1. (a) Basic idea behind a single-bit CTDSM with an FIR feedback and (b) a semi-digital realization of the FIR DAC.

loop filter. Our analysis and simulations show that the sensitivity of an FIR DAC based CTDSM with respect to RC spread is similar to that of a conventional design. Another issue is the choice of the number of taps in the FIR filter. At first glance, it appears as if increasing the number of taps must result in improved performance (as the shaped quantization noise is more heavily filtered). In this work, we show that there is little (if any) benefit as the number of taps is increased beyond a certain value. Rise-fall asymmetry of the feedback DAC waveform causes inter symbol interference, which manifests as even order distortion and an elevated in-band noise floor at high input amplitudes. This has to be considered during the design phase, in a high resolution modulator such as this.

This paper, which is a more detailed version of [9], presents an FIR DAC based single-bit CTDSM that addresses these issues. The modulator, designed in a 0.18 μm CMOS process, achieves a dynamic range of 103 dB and an SFDR (spurious free dynamic range) of 106 dB in a 24 kHz bandwidth while dissipating only 280 μW from a 1.8 V supply. The test chip also includes a dither generator, DAC rise-fall asymmetry correction circuitry, decimator and an SPI interface (serial peripheral interface). The decimator, which filters out of band quantization noise, and yields a 20 bit 48 KS/s sequence, consumes 100 μW . The design of the chip forms the subject of the rest of the paper, which is organized as follows. In Section II, we describe the architecture and develop the stabilization technique that compensates the loop for the delay introduced by the FIR DAC. We give intuition for the optimal number of FIR filter taps for low power operation. Section III discusses circuit design details. Experimental results and comparison with state of the art designs are given in Section IV. Section V concludes the paper.

II. ARCHITECTURAL DEVELOPMENT

The architecture of the prototype modulator on which the FIR-CTDSM is based is a single loop design as shown in Fig. 2(a). The sampling rate is assumed to be 1 Hz. The third order loop filter is realized as a CIFF-B (cascade of integrators feedforward and feedback) structure [10]. This architecture has several advantages. The fast path around the quantizer (through DAC_2 and I_1) and the high gain path (through I_1 , I_2 and I_3) can be independently optimized, like in a CIFB (cascade of integrators feed-

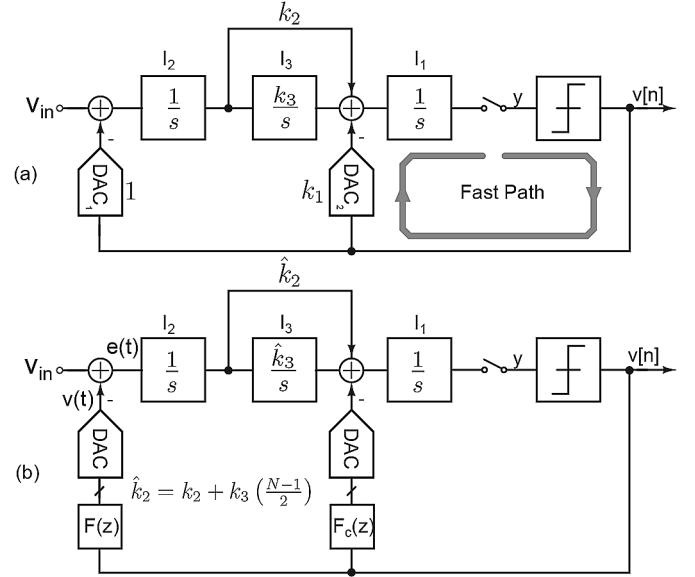


Fig. 2. (a) Prototype CIFF-B modulator and (b) FIR-DAC CTDSM based on the prototype.

TABLE I
DISCRETE-TIME TRANSFER FUNCTIONS FOR EACH PATH THROUGH THE LOOP FILTER, ASSUMING AN NRZ DAC PULSE SHAPE

Path	DT Transfer Function
$\frac{1}{s}$	$\frac{z^{-1}}{1 - z^{-1}}$
$\frac{1}{s^2}$	$\frac{0.5(z^{-1} + z^{-2})}{(1 - z^{-1})^2}$
$\frac{1}{s^3}$	$\frac{0.17z^{-1} + 0.66z^{-2} + 0.17z^{-3}}{(1 - z^{-1})^3}$

back [11]) design. At high frequencies, the STF (signal transfer function) rolls off as $1/s^2$. Due to feedforward, the output of I_2 has virtually no signal at the input frequency. This means that its gain in the signal band (after dynamic range scaling) will be large. Thanks to this, non idealities in the rest of the loop filter will be significantly attenuated when referred to the modulator input, just like in a CIFF (cascade of integrators with feedforward) design. The CIFF-B architecture, therefore, inherits the appealing aspects of both its parents. The gain of the loop filter from v to y is $k_1/s + k_2/s^2 + k_3/s^3$. The coefficients are chosen so that the resulting NTF is maximally flat (assuming an NRZ (non return to zero) DAC) and has an out-of-band gain of 1.5 in accordance with Lee's rule [12].

The z -domain transfer functions for the $1/s$, $1/s^2$ and $1/s^3$ paths, which are a ratio of polynomials in z^{-1} , and denoted by $H_1(z)$, $H_2(z)$ and $H_3(z)$ respectively, are given in Table I. An NRZ pulse shape is assumed for the DAC. The numerators of these transfer functions, denoted by $N_1(z)$, $N_2(z)$ and $N_3(z)$, all evaluate to unity for $z = 1$. The coefficients k_1 , k_2 and k_3 can be determined by solving

$$k_1 H_1(z) + k_2 H_2(z) + k_3 H_3(z) = \frac{1}{NTF(z)} - 1. \quad (1)$$

The main feedback DAC (DAC_1) in the prototype is then replaced by an N -tap FIR DAC, whose transfer function is de-

noted by $F(z)$, as shown in Fig. 2(b). All taps of the FIR filter are assumed to be identical for ease of layout. Further the DC gain of $F(z)$ should be 1, to ensure that the in-band STF of the CTDSM is unity. Thus,

$$F(z) = \frac{1}{N}(1 + z^{-1} + \dots + z^{-(N-1)}) = \frac{(1 - z^{-N})}{N(1 - z^{-1})}. \quad (2)$$

To stabilize the loop so that the NTF is restored, a compensation FIR DAC with transfer function $F_c(z)$ is added at the input of I_1 , and the loop filter coefficients are tuned, as shown in Fig. 2(b). This means that the gain from v to y in both designs should be the same at all (discrete-time) frequencies. In [9], only the final results of the compensation process, enumerated below, were given due to paucity of space.

- a) $\hat{k}_2 = k_2 + (k_3/2)(N - 1)$;
- b) $F_c(z)$ is an N -tap FIR filter.

Here, we derive these results. The intuition for $\hat{k}_2$ (which is larger than k_2) and $F_c(z)$ is the following. The main FIR DAC, due to its lowpass nature, reduces the high frequency contributions of the $1/s^3$ and $1/s^2$ paths to the loop gain of the modulator. To restore the loop gain (and thereby the NTF), the gain at high frequency has to be increased. This is accomplished by increasing k_2 and through another FIR filter $F_c(z)$ which excites the $1/s$ path.

In the discussion to follow, all hatted quantities refer to the modulator with the FIR DAC. Equating the loop gains of both the modulators in Fig. 2, we have

$$\hat{k}_3 F(z) H_3(z) + \hat{k}_2 F(z) H_2(z) + F_c(z) H_1(z) = k_3 H_3(z) + k_2 H_2(z) + k_1 H_1(z). \quad (3)$$

At very low frequencies (i.e. $z \rightarrow 1$), the loop gain of both CTDSMs is dominated by the third order path. Equating these gains, and recognizing that $F(1) = 1$, we obtain

$$\hat{k}_3 F(z) H_3(z) \Big|_{z \rightarrow 1} = k_3 H_3(z) \Big|_{z \rightarrow 1} \Rightarrow \hat{k}_3 = k_3. \quad (4)$$

Using $\hat{k}_3 = k_3$ in (3), and moving $F_c(z) H_1(z)$ to the right hand side, we have

$$\hat{k}_2 F(z) H_2(z) = k_3 [1 - F(z)] H_3(z) + k_2 H_2(z) + (k_1 - F_c(z)) H_1(z). \quad (5)$$

From the equation above, we see that $\hat{k}_2$ is given by

$$\hat{k}_2 = \lim_{z \rightarrow 1} \frac{k_3 [1 - F(z)] H_3(z) + k_2 H_2(z) + (k_1 - F_c(z)) H_1(z)}{F(z) H_2(z)}. \quad (6)$$

Using Table I for $H_1(z)$, $H_2(z)$ and $H_3(z)$, and observing that $F(1) = 1$, we have

$$\hat{k}_2 = \lim_{z \rightarrow 1} \frac{k_3 [1 - F(z)]}{1 - z^{-1}} + k_2 + \lim_{z \rightarrow 1} (k_1 - F_c(z)) (1 - z^{-1}). \quad (7)$$

A useful observation to make regarding $[1 - F(z)]$ is the following. Since $F(1) = 1$, it follows that $[1 - F(z)]$ has a zero at $z = 1$, enabling it to be expressed as

$$1 - F(z) = M_1(z)(1 - z^{-1}) \quad (8)$$

where $M_1(z)$ is a polynomial of order $(N - 2)$ in z^{-1} . Further, using (2) and l'Hopital's rule, we see that

$$M_1(1) = \lim_{z \rightarrow 1} \frac{1 - F(z)}{1 - z^{-1}} = \frac{1}{2}(N - 1) \quad (9)$$

and (7) yields

$$\hat{k}_2 = k_2 + k_3 M_1(1) = k_2 + \frac{k_3}{2}(N - 1). \quad (10)$$

Referring to (3), $F_c(z)$ is the only remaining unknown. Using $\hat{k}_2 = k_2 + k_3 M_1(1)$, and $(1 - F(z)) = M_1(z)(1 - z^{-1})$, (3) can be recast as follows

$$F_c(z) = k_1 + k_2 \frac{N_2(z) M_1(z)}{H_1(z)(1 - z^{-1})} + k_3 \frac{M_1(z) N_3(z) - M_1(1) F(z) N_2(z)}{H_1(z)(1 - z^{-1})^2} \quad (11)$$

where N_1 , N_2 and N_3 denote the numerator polynomials of H_1 , H_2 and H_3 respectively. Since $F(z)$, $N_3(z)$ and $N_2(z)$ evaluate to 1 for $z = 1$, it is easy to see that $M_1(z) N_3(z) - M_1(1) F(z) N_2(z)$ has a zero at $z = 1$. This enables us to write

$$M_1(z) N_3(z) - M_1(1) F(z) N_2(z) = M_2(z)(1 - z^{-1}) \quad (12)$$

where $M_2(z)$ is an N^{th} order polynomial in z^{-1} . Using this, and $H_1(z) = z^{-1}/(1 - z^{-1})$, (11) simplifies to

$$F_c(z) = k_1 + k_2 z N_2(z) M_1(z) + k_3 z M_2(z) \quad (13)$$

from which it is immediately apparent that $F_c(z)$ has to be an N tap FIR filter. The theory above establishes that by tuning k_2 , and adding $F_c(z)$ at the input of I_1 , the NTF can be restored *exactly*. Once this recognized, the coefficients of F_c can also be determined either analytically as shown above, or by using numerical techniques, by proceeding along lines similar to [13]. Fig. 3 shows the magnitude responses of the first, second and third order paths in a conventional modulator and that with FIR feedback. In the latter, the loop is assumed to have been compensated in accordance with the discussion above. From part (b) of the figure, it is apparent that the second and third order paths have reduced high frequency content due to the FIR filter. To compensate for this loss, the gain of the first order path has to be increased (in a frequency dependent way), and is accomplished by the compensating filter $F_c(z)$.

A. Choice of Number of Taps in the FIR DAC

It is tempting to increase the number of taps in the main FIR DAC without limit, since a longer FIR DAC results in better filtering of the quantization noise. Due to this, one would tend to conclude that the magnitude of the error signal $e(t)$ (in Fig. 2(b)) would reduce as N increases. This is true for small N – better filtering of the quantization noise does indeed reduce $|e(t)|$.

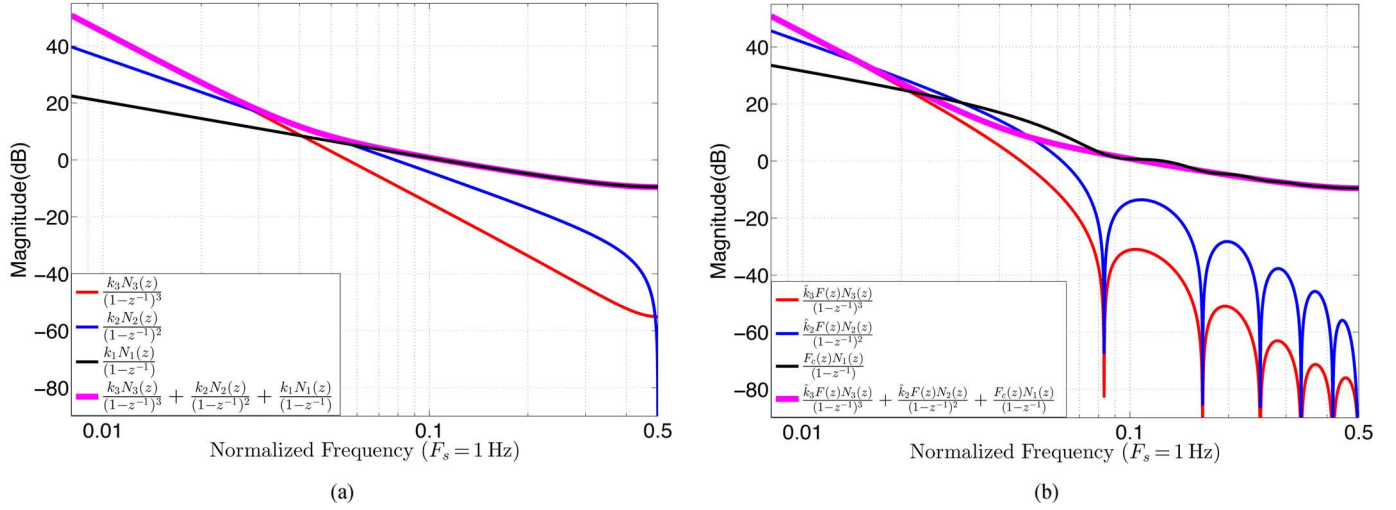


Fig. 3. Frequency responses of the first, second and third order paths of the loop gain function in (a) a single-bit modulator and (b) a modulator with a 12-tap FIR DAC. The coefficients are tuned and the compensation filter is chosen so that the NTF is restored.

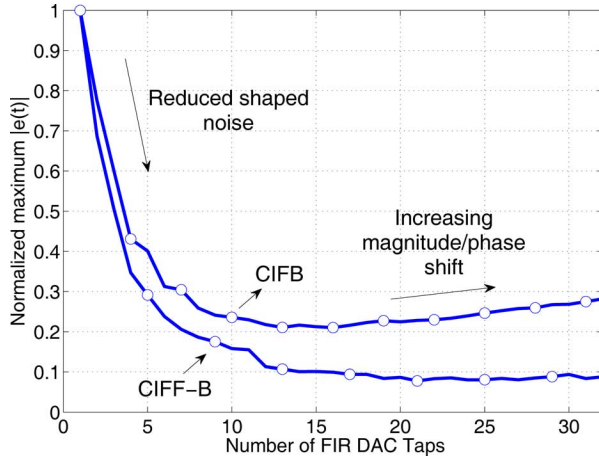


Fig. 4. Peak magnitude of the error signal exciting the input integrator versus the number of taps in the main FIR DAC, for CIFB and CIFF-B loop filters.

However, beyond a certain point, $|e(t)|$ no longer decreases due to the following.

The z -transform of CTDSM output sequence is given by $V(z) = NTF(z)Q(z) + STF(z)V_{in}(z)$, where V_{in} and Q denote the transforms of the sampled input (assumed to be an inband signal) and the quantization noise respectively. The error sequence $V_{in}(z) - F(z)V(z)$ is given by $E(z) = \mathcal{Z}[e[n]] = -Q(z)NTF(z)F(z) + V_{in}(z)(1 - F(z)STF(z))$, where $F(z)$ is the (lowpass) transfer function of the FIR DAC. The waveform processed by the loop filter is $e(t) = \sum_k e[k]p(t - kT_s)$, where $p(t)$ denotes the DAC pulse shape. The peak magnitude of the error waveform, therefore, is influenced by two conflicting factors.

- Shaped quantization noise, filtered by the FIR DAC.
- An input component, proportional to the deviation of the STF from unity.

As the number of taps (N) in the FIR DAC is increased, better filtering of the shaped noise causes $|e(t)|$ to reduce. On the other hand, larger N increases $(1 - F(z)STF(z))$ due to the following. As seen from (10), k_2 increases with N . Further,

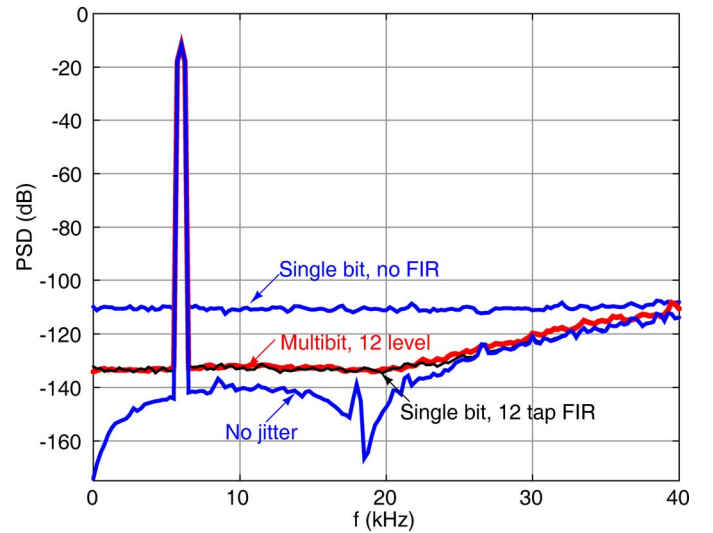


Fig. 5. A comparison of the effect of clock jitter for several CTDSM design choices.

going through the calculations in (13) reveals that the DC gain of $F_c(z)$ increases with N . Both of these increase STF peaking (see Fig. 8), causing the input component of the fed back signal $v(t)$ to be larger in magnitude and shifted in phase with respect to $v_{in}(t)$. This means that even though the quantization noise component of $v(t)$ is smaller, $|e(t)|$ is larger even though N is increased. The increased DC gain of $F_c(z)$ is also problematic in a practical implementation, as the input signal component injected by the compensation DAC necessitates a lower unity gain frequency for I_3 (after dynamic range scaling). Further, the power dissipation of the clock generation and distribution circuitry increases with the number of taps.

The “optimal” number of taps to be used in the FIR DAC, therefore, is dependent on the loop filter topology (which influences the STF) and the input signal frequency. In a CIFF-B design, it is a trade off between the amount of STF peaking one is willing to tolerate, the increased power dissipation due

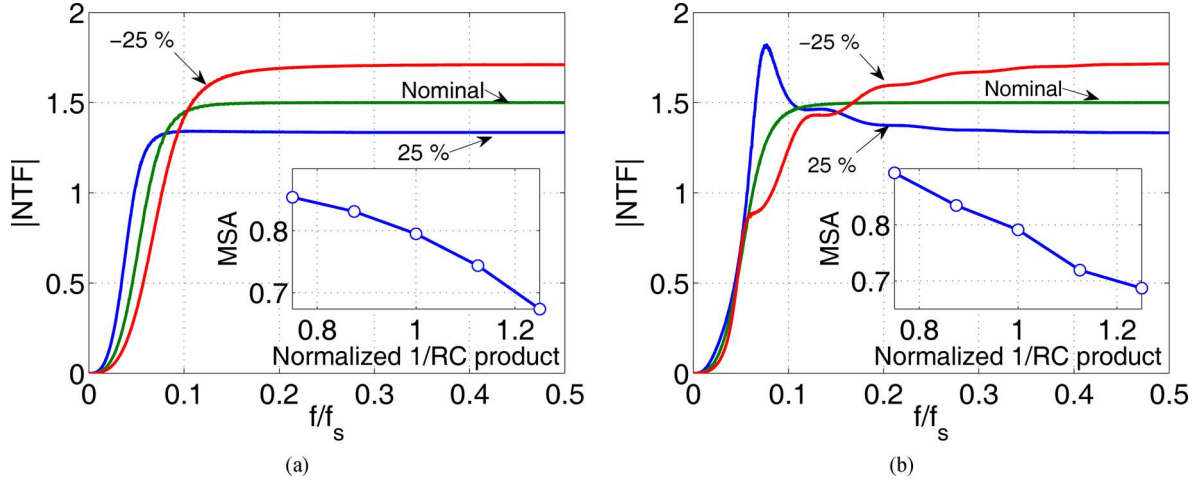


Fig. 6. Effect of RC time constant variations on a CTDSM with (a) an NRZ DAC (b) 12-tap FIR DAC. The insets show the maximum stable amplitude (MSA) as time constants deviate $\pm 25\%$ from their nominal values.

to the extra taps (without a corresponding decrease in $|e(t)|$) and the DC gain of $F_c(z)$, which has implications for the design of I_3 . Simulation results of $e(t)$ (normalized to the value that would be obtained without an FIR filter), as shown in Fig. 4 for CIFB and CIFF-B loop filters were used as a guide to decide that 12-tap FIR DACs represented a reasonable choice considering the trade offs involved.

B. Effect of Clock Jitter

It is well known that clock jitter degrades the performance of a CTDSM by introducing errors in the waveform of the feedback DAC. The error due to jitter in each clock cycle is proportional to the height of the transition of the DAC waveform at the beginning of the cycle. Since a single-bit modulator results in a rail-to-rail feedback waveform, one should expect a severe degradation of the inband SNR in the presence of clock jitter. The use of a single-bit quantizer + FIR DAC, or a multibit quantizer – can both address this problem. Fig. 5 compares the jitter sensitivity of several CTDSM design choices. The signal bandwidth is 24 kHz. White clock jitter, with an RMS value of 50 ps is assumed. The single-bit modulator has an OSR of 128. All taps of the FIR DAC are equal. For the multibit case, a 12-level quantizer, and an NTF with an out-of-band gain of 2.5 is used. Due to multibit operation, the OSR is reduced to 42, so as to result in the same SQNR as in the single-bit design. As seen from Fig. 5, using an FIR DAC suppresses jitter noise of a single-bit modulator by about $20 \log(12) \approx 21$ dB, and has the same performance as a 12-level multibit CTDSM operating at a third of the sampling frequency. The ideal spectrum (without jitter) is also shown for comparison.

C. Effect of Time Constant Variations

A concern regarding an FIR DAC based CTDSM is the effect of time constant variations on the NTF, and the maximum stable amplitude (MSA). To examine this, simulations were run on CTDSMs with and without an FIR DAC. Under nominal conditions, both modulators were designed to have the same maximally flat NTF with an out of band gain of 1.5. All time constants were then changed over a $\pm 25\%$ range from their nominal values.

The resulting NTFs and MSAs for the conventional and FIR-CTDSM are shown in Fig. 6(a) and (b) respectively. It is seen that the out of band gains are similar in both cases. Since the high frequency gain of the NTF largely influences the MSA, it makes sense that both designs have similar MSAs as time constants are varied. From this, we conclude that using an FIR-CTDSM is not any more sensitive to systematic time constant variations than its NRZ counterpart.

III. CIRCUIT DESIGN

A. Modulator Architecture

A simplified single-ended schematic of the third order CTDSM is shown in Fig. 7(a). As discussed in the preceding section, a CIFF-B loop filter is used. Negative resistors indicate inversion in the differential version. Active-RC integrators are used for low noise and high linearity. FIR DACs inject currents proportional to the reference into the virtual grounds of the opamps. R_f feeds the input into the third integrator formed by A_3 - R_3 - C_3 . Without R_f , the output of A_2 would consist of a component proportional to V_{in} . This is due to the following. The magnitude of the low frequency current in C_3 must be ≈ 0 . This means that the sum of the low frequency currents due to the compensating FIR DAC ($FIR_c(z)$) and the voltage at the output of A_2 must be ≈ 0 , indicating that the second integrator must “absorb” the low frequency output of $F_c(z)$. To avoid saturation of this integrator, its integrating capacitor must be large. Using R_f to cancel the low frequency output of $F_c(z)$ (which is largely proportional to V_{in}) avoids this problem, enabling the use of a much smaller C_2 .

The NTF of the modulator has optimized zeros for reduced in-band quantization noise. Since this is an audio design, the frequency of the NTF zeros is small (about 18 kHz), necessitating the realization of large time constants. In our design, the complex zeros of the NTF are implemented using weak feedback around the second and third integrators, which are comprised of A_2 - R_2 - C_2 and A_3 - R_3 - C_3 , as explained below. The T-network formed by R_x , R_y and R_z inject a current proportional to the output of A_3 into the virtual ground node of A_2 , as shown in Fig. 7(a). Since $R_y \ll R_x, R_z$, the current flowing through

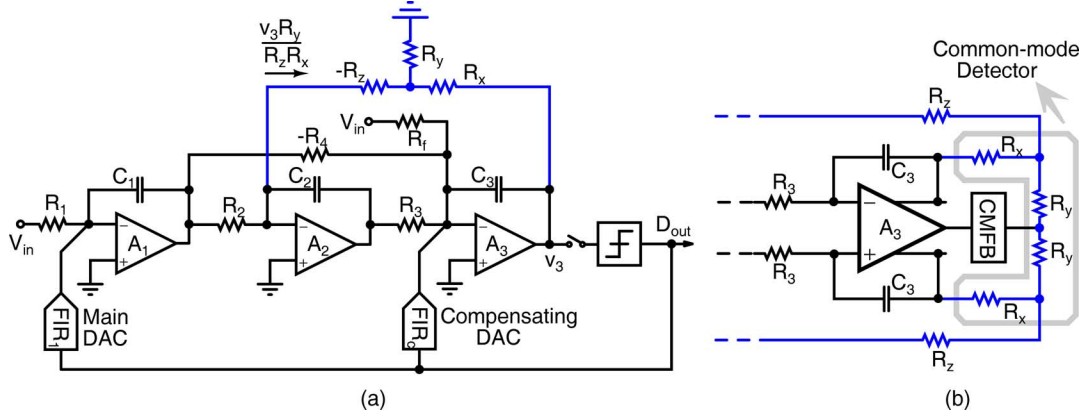


Fig. 7. (a) Simplified single-ended schematic of the CTDSM and (b) reusing the common-mode detector at the output of A_3 to realize R_x and R_y .

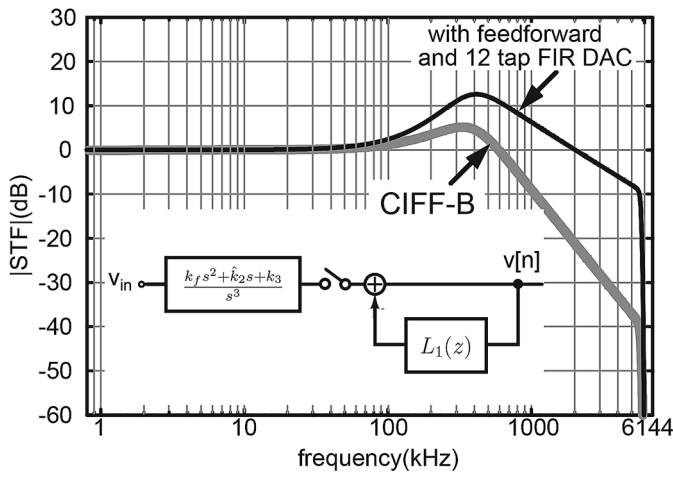


Fig. 8. Simulated STFs of a CIFF-B modulator with an NRZ DAC, and the same modulator with a 12-tap FIR DAC (and a corresponding compensation DAC) and input feedforward, as in Fig. 7.

R_z is given approximately by $V_3 (R_y/R_x) (1/R_z)$. Without the T-network, R_z would have to increase by a factor R_x/R_y , requiring a lot of area. In the differential implementation, R_z taps an appropriate point of the resistive common-mode detector, needed anyway, at the output of A_3 , as shown in Fig. 7(b). Thus, no extra resistors are needed to realize R_y and R_x [14].

1) *Effect of the FIR DAC and Input Feedforward on the Signal Transfer Function:* The FIR DAC and the compensating filter $F_c(z)$, as well as input feedforward (through R_f in Fig. 7) modify the STF of the modulator. The STF can be derived by separating the modulator into continuous-time and discrete-time parts [15], as shown in the inset of Fig. 8. In this figure, $L_1(z)$ denotes the equivalent discrete-time transfer function from D_{out} to v_3 in Fig. 7. The STF of the modulator is seen to be

$$STF(j\omega) = \frac{k_f s^2 + \hat{k}_2 s + k_3}{s^3} \Big|_{s=j\omega} NTF(e^{j\omega}) \quad (14)$$

where $NTF(z) = 1/(1 + L_1(z))$. For a CIFF-B modulator without feedforward or an FIR DAC, the STF would be that in the above equation with $k_f = 0$ and $\hat{k}_2 = k_2$. From the discussion in Section II, $\hat{k}_2 = k_2 + k_3(N-1)/2$, indicating that one should expect an increased peaking in the signal transfer function with respect to a CIFF-B CTDSM without an FIR DAC

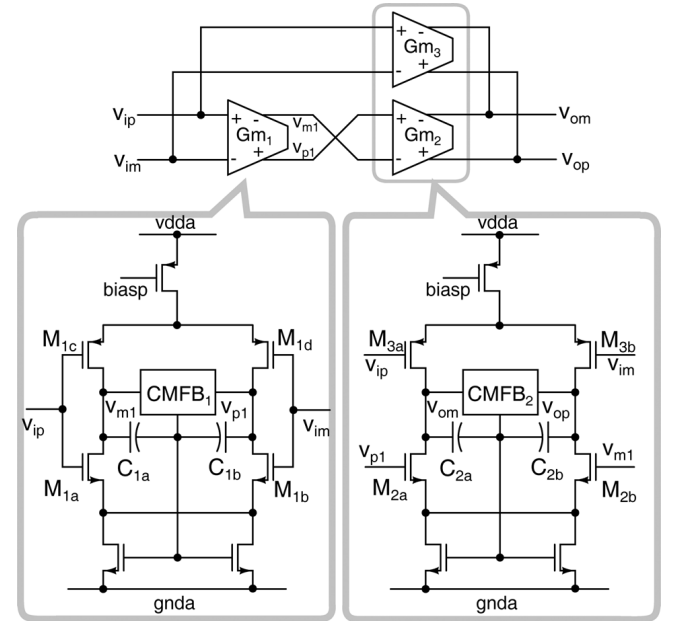


Fig. 9. Simplified schematic of the first operational amplifier.

and feedforward. This is indeed seen in simulation results, as is apparent from the plots of Fig. 8. The NTF is not effected by input feedforward.

All integrating capacitors are realized as switchable banks, to counter RC time constant variations. The opamps are two stage feedforward compensated structures. Thanks to the low frequency of operation (in relation to the speed of the process), excess loop delay is negligible. For low noise operation, the 12-tap FIR DACs are implemented using resistors. Since the DAC is realized as a semi-digital structure [16], they are inherently linear: resistor mismatch only alters the transfer function of the FIR filter, which is virtually inconsequential to modulator performance. Unfortunately, rise-fall asymmetry can cause even order distortion due to inter symbol interference (ISI), like in an a conventional NRZ DAC. Use of FIR filtering *does not* mitigate this problem [6]. Since reliable matching data for this process was not available, an ISI tuning loop that equalizes the rise and fall times of the DAC pulse is implemented on-chip. For simplicity of layout, all taps in the main FIR DAC (FIR_1) are made identical. Details of the individual blocks are given below.

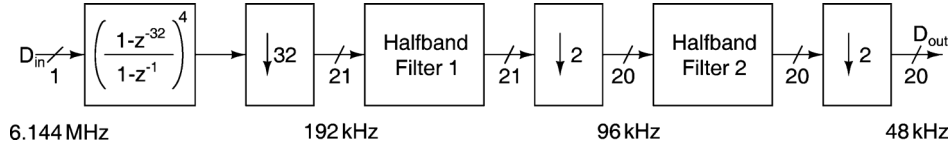


Fig. 12. Block diagram of the decimation filter.

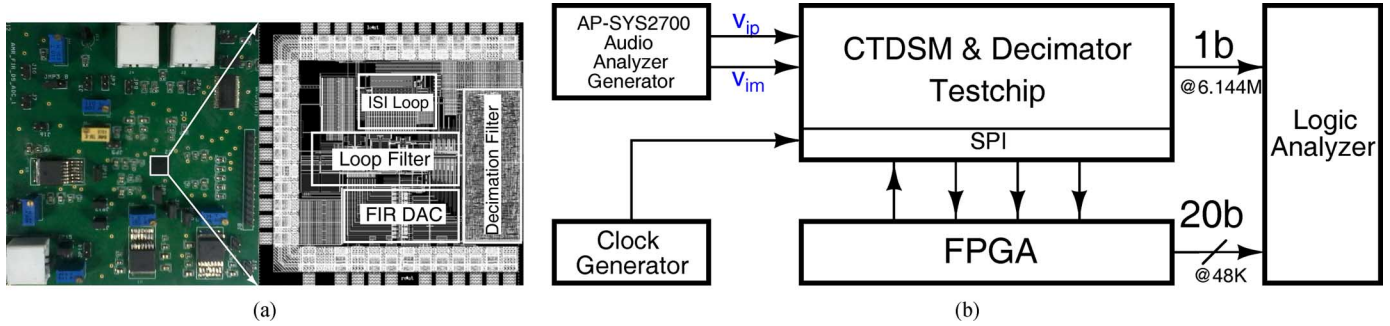


Fig. 13. (a) Photograph of the testboard and layout and (b) schematic of the test setup.

in multiple stages as shown in Fig. 12. The first stage is a fourth order 32-tap CIC filter, implemented as a Hogenauer structure [18]. This is followed by two halfband FIR low pass filters, each of whose outputs are downsampled by a factor of 2. The filter orders are 16 and 60 respectively. The twenty bit decimation filter output is transferred outside through an SPI interface, which also serves to load tuning bits needed in the design.

F. Comparison With A Multibit Modulator Targeting the Same Specifications

It is worthwhile to compare the power dissipation of a multibit modulator with that of a single-bit (+FIR DAC) CTDSM designed in the same process. For this purpose, we use a 12-level quantizer operating at a third of the sampling rate. Each sub-block of a CTDSM can be implemented in several ways, making it difficult to accurately assess the exact benefit of using a single-bit quantizer. The comparison that follows is an estimate based on our experience with the multibit design of [19].

- a) **ADC** : A 12-level quantizer needs 11 comparators. Due to the reduced sampling rate, the dynamic power is reduced by a factor of 3. However, random offsets in the comparators degrade inband noise performance, as seen from simulation results given in [19, Fig.12]. These results indicate that the standard deviation of the comparator random offset should be restricted to about 0.1 LSB to achieve a worst case SNDR of 105 dB. This means that the power efficient comparator used in the single-bit design is not appropriate in a multibit scenario. Apart from the reference ladder, an offset calibration loop will be necessary to reduce SNDR degradation due to random offsets in the flash ADC. Another aspect that is usually not considered during the design phase is the power needed in the clock generator, which supplies the clock phases needed for the operation of the comparators. Experience shows that this can draw significant current. Our estimate, based on simulations of [19] is that the ADC will need to consume at least $50 \mu\text{W}$ in this technology, as compared to $4 \mu\text{W}$ in the single-bit case (including the clock generator).

- b) **DAC and DEM logic**: Dynamic Element Matching (DEM) logic, needed in a multibit design, is not needed in the FIR-DAC case. Further, at high resolutions such as this, inter symbol interference (ISI) in the unit elements of the DAC can degrade performance even with DWA [20]. This will necessitate an ISI tuning loop like the one used in our work, or a more complex DEM algorithm which digitally shapes out ISI as in [20]. Our estimate is that, in spite of the increased sampling rate, the power consumed by the digital portion of the FIR DAC (about $20 \mu\text{W}$) will be about the same as that in a multibit DAC (whose linearity is obtained through DEM).
- c) **Loop Filter**: The output swing of the loop filter in the case of a multibit modulator has to be as large as possible so as to relax the offset requirements on the flash comparators. This dictates the use of an opamp with a peak-to-peak swing approaching the supply rails. This is most easily possible with a Miller compensated opamp. Further, the load presented by the flash ADC is bound to be at least 20 times (considering increased routing parasitics) that presented by the single-bit ADC. From this discussion, it appears as if the opamps will burn considerably more power when compared to a single-bit design, where these constraints do not exist.

From the arguments above, it is apparent that the use of a single-bit modulator with an FIR DAC saves power on all fronts, while achieving similar (or marginally better) performance. This is further borne out by our experimental results and comparison with state of the art modulators, presented in the following section.

IV. MEASUREMENT RESULTS

The CTDSM, decimator and an SPI interface (to facilitate data transfer to/from the chip) were fabricated in a $0.18 \mu\text{m}$ CMOS process through the Europractice program. Fig. 13(a) shows the layout and the test board used for characterization. The active area of the chip, including the decimator, is $1.25 \text{ mm} \times 1 \text{ mm}$.

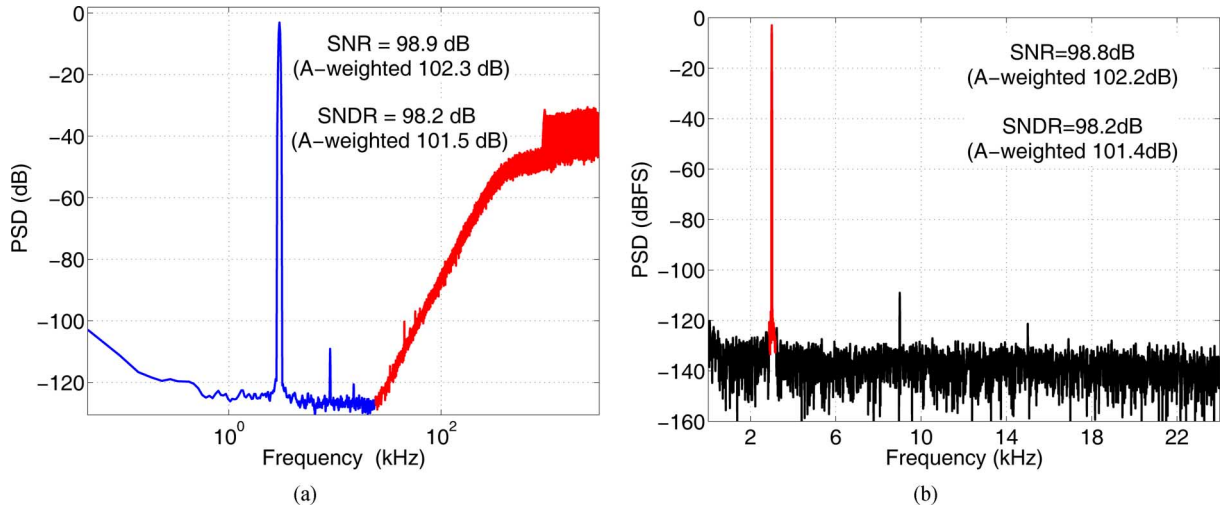


Fig. 14. (a) PSD of the CTDSM output at peak SNDR and (b) PSD of the decimated output.

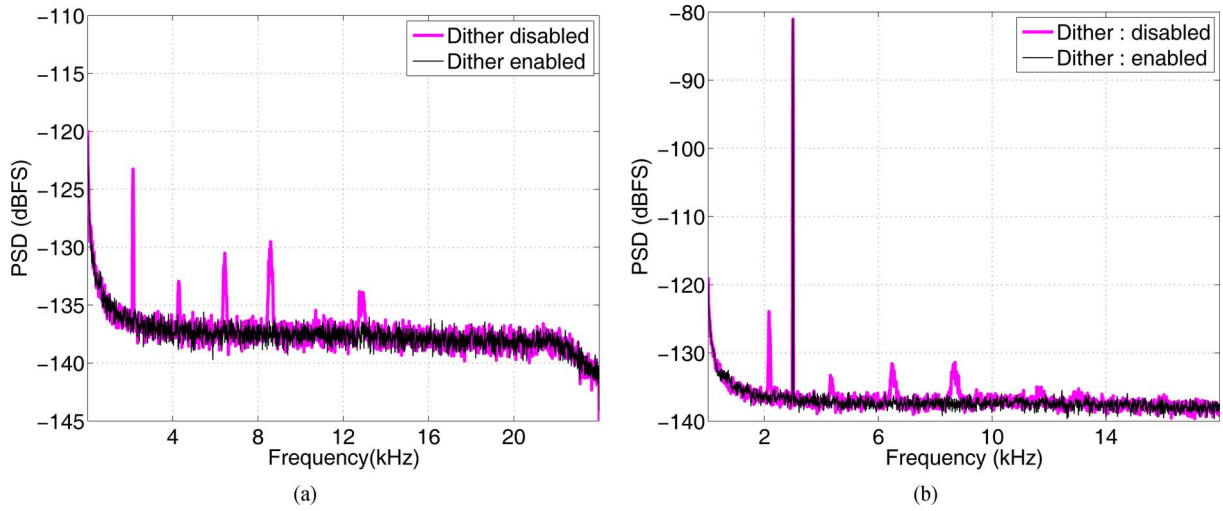


Fig. 15. (a) PSD under idle channel conditions with(out) dither and (b) PSD for a -80 dBFS input tone (with)out dither.

The test setup is shown in Fig. 13(b). An Audio Precision AP-SYS 2700 was used as the input source. The test chip had a provision to enable capture of either the single-bit output of the modulator at 6.144 MS/s, or the decimated output at 48 KS/s through an FPGA. Fig. 14(a) shows the PSD of the modulator at the amplitude that results in the peak SNDR. The peak SNR, SNDR and dynamic range (DR) are 98.9 dB, 98.2 dB, and 103 dB respectively. The peak A-weighted SNR and SNDR are 102.3 dB and 101.5 dB respectively. The third harmonic distortion is seen to be about -106 dB. The PSD computed at the output of the decimation filter indicates virtually the same performance, demonstrating the performance of the decimator.

Fig. 15(a) and (b) show the PSD of the decimated output, with and without dither enabled, for an idle channel and -80 dBFS input respectively. It is seen that dither has successfully eliminated idle tones. Fig. 16 demonstrates the efficacy of the proposed ISI calibration scheme. When ISI correction is disabled, second harmonic distortion is seen to be about -121 dB. With correction, the second harmonic is buried in the noise floor. The inherently low HD_2 (even without correction) indicates good resistor matching in this process. Fig. 17 shows the plot of

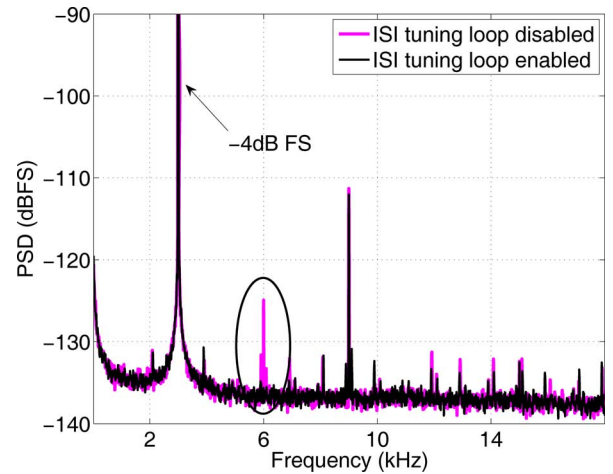


Fig. 16. PSD of the decimated output with(out) ISI tuning, for a -4 dBFS input tone. The second harmonic is largely eliminated.

SN(D)R as a function of input amplitude. Table II shows a summary of measured performance. Table III compares this work

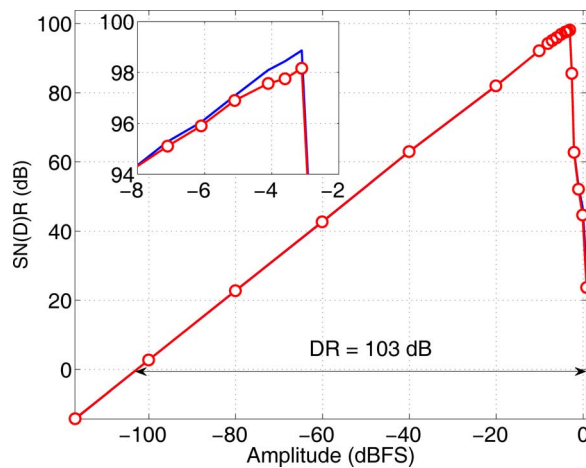


Fig. 17. Measured SNR and SNDR. They are indistinguishable except at high amplitudes. The peak SNR, SNDR and Dynamic Range are 98.9 dB, 98.2 dB and 103 dB respectively.

TABLE II
SUMMARY OF MEASURED PERFORMANCE

Signal bandwidth/Clock rate	24 kHz/6.144 MHz
DR/SNR/SNDR	103 dB/98.9 dB/98.2 dB
A-weighted SNR/SNDR	102.3 dB/101.5 dB
SFDR	106 dB
Process/Supply/Quantizer range	180 nm CMOS/1.8 V/3.6 $V_{pp,diff}$
Power dissipation(Modulator+references)	280 μ W
Power dissipation(Decimation filter)	100 μ W
FoM _{SNDR} /FoM _{Schreier}	88 $\frac{fJ}{Vl}$ / 182.3 dB

$$FoM_{SNDR} = \frac{Power_{SNDR=176}}{2 \cdot BW \cdot 2^{176/6.02}} \quad FoM_{Schreier} = DR + 10 \log \left(\frac{BW}{Power} \right)$$

TABLE III
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART AUDIO $\Delta\Sigma$ CONVERTERS

Reference	BW (kHz)	SNDR (dB)	SFDR (dB)	Power (μ W)	Feature Size (nm)	Supply (V)	FoM (fJ/level)	Schreier FoM
[1]	20	95	-	37000	180	3.3	20122	158.3
[2]	20	97	116	18000	330	3.3	7776	165.45
[3]	20	94	96	15000	65	3.3	9153	162
[21]	20	97.5	102	9900	130	3.3	4037	168
[22]	25	95	106	870	180	0.7	379	174.5
[23]	20	99.3	112	1100	180	5	365	173.9
[24]	24	95	103.4	371	65	1	168	174.5
[19]	24	90.8	94	121	180	1.8	89	176.5
[14]	24	88	95	110	180	1.8	111	175.9
[25]	25	70	-	135	500	1.5	1040	162.7
[26]	20	73.1	-	60	130	0.9	406	168.2
This work	24	98.2	106	280	180	1.8	88	182.3

with state of the art designs in the literature. [1]–[3], [21]–[24] achieve SNDRs greater than 94 dB – they all employ *multibit quantizers*. Our CTDSM achieves significantly better FoM and a higher SNDR (excepting [23]), in spite of using a single-bit architecture. [19] and [14] are audio CTDSMs targeting similar specifications in the same process technology. It is interesting to compare them with the modulator presented in this work, as both were designed in our group. [19] is a multibit design, while [14] is a single-bit CTDSM with the linearity of the first opamp being enhanced by the assisted opamp technique. It is seen that the modulator presented in this paper has an FoM better than that of [19], a 12 dB higher SFDR, and achieves a higher SNDR at the same time. This demonstrates the inherent architectural advantage of using FIR feedback.

V. CONCLUSIONS

We demonstrated that single-bit CTDSMs can achieve very high linearity in a power efficient manner by using an FIR feedback DAC. A technique for compensating the $\Delta\Sigma$ loop for the delay of the FIR DAC, thereby restoring the NTF of the modulator, was described. We showed that the number of taps in the FIR DAC should be chosen based on the modulator architecture and the OSR. Measurement results from a test chip designed in a 0.18 μ m CMOS technology, which included a decimator in addition to the modulator, showed that the modulator achieves a peak A-weighted SNR of 102.3 dB, a peak raw SNDR of 98.9 dB and an SFDR of 106 dB while dissipating 280 μ W from a 1.8 V supply. This performance translates to an FoM of 88 fJ/level ($FoM_{Schreier} = 182.3$ dB). The theory and techniques described in this paper are, therefore, borne out by our test results.

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